Teacher's Signature:

Experiment No. 4

Dated : 8/8/25

Objective : Arithematic Progression: To verify sum of first n natural numbers

## Pre-requisite Knowledge:

* Arithematic Progression

Counting no.s 1, 2, 3 ---- are natural nos. Smallest natural no. is 1 while largest no. cannot be found.

* $n^{th}$ term of an AP

Let $a, a+d, a+2d$ ---- be an AP, then $n^{th}$ term is - $T_n = a + (n-1)d$

where 'a' is first term and 'd' common.

* Sum of n terms of a general AP

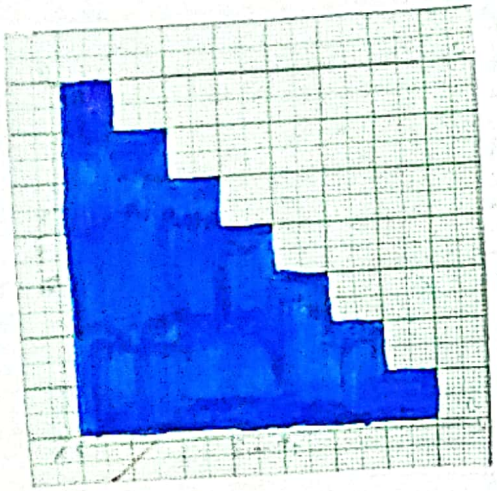
$$S_n = \frac{n}{2}[2a + (n-1)d]$$

* Sum of first n natural nos.

$$S_n = \frac{n}{2}[2 \times 1 (n-1)] = n(n-1)$$

## Material Required:

Colour papers, graph paper (cm), stretch pen, scissors, glue, pen



Teacher's Signature: 12/8

Experiment No. .....

Objective : .....

Dated : .....

- **Demonstration:**

1. The shaded region, looks like stair case.
2. Length if 1st stair is 1 unit, length of 2nd stair is 2 units and so on length of 7th stair is 7 units.
3. Sum of first 7 terms $= 1+2+3 - \cdots + 7 = 28$ — (i)
4. We observe that,

$$\text{area of region} = \frac{1}{2} \text{ area of } ABCD$$

$$= \frac{1}{2} \times l \times b$$

$$= \frac{1}{2} \times 7 \times 8 = 28$$

$\therefore$ This result can be generalised to find sum of $n$ natural nos. as $S_n = \frac{1}{2} n(n+1)$ — (ii)

- **Observation:**

For $n = 4$, in eq 2, $S_n = 10$

For $n = 9$, in eq 2, $S_n = 45$

For $n = 40$, in eq 2, $S_n = 1440$

For $n = 100$, in eq 2, $S_n = 5050$

- **Result:**

We have found $S_n = \frac{1}{2} n(n+1)$

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